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(U) ShoreECDIS Software User Manual (SUM)

(U) Version 2.1.0

Prepared For:	**Prepared By:**
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(U) Change Log

Version	Date	Items Affected	Remarks
1.0.0	11 JUL 2023	Initial Document	Initial release of ShoreECDIS User Manual (v2.1.0)
2.0.0	Current	Compatibility & Standard Specs	Updated specification support matrix for S-100 v2.0.0 standards
2.1.0	Current	UI, Projection, Decryption, GPS, Scales	Added temporal inputs, multiple projections, GPS configurations, decryption keys, SENC backwards-compatibility updates, custom scales, color palettes, and viewing group controls

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1. (U) Scope

1.1 (U) Identification

This document applies to the **ShoreECDIS** software application developed by Naval Information Warfare Center Atlantic (NIWC-LANT).

- **Software Name:** ShoreECDIS
- **Document Title:** ShoreECDIS Software User Manual (SUM)
- **Current Software Version:** 2.1.0

1.2 (U) System Overview

ShoreECDIS is an Electronic Chart Display and Information System (ECDIS) testbed application created to demonstrate that traditional operational ECDIS requirements can be met using modern International Hydrographic Organization (IHO) S-100 universal hydrographic data model standards. The software serves hydrographic offices, mariners, and testbed evaluation teams by rendering S-100 vector and raster datasets (e.g., S-101 Electronic Navigational Charts, S-102 Bathymetric Surface data, S-104 Water Level information, S-111 Surface Currents, S-124 Navigational Warnings, and S-129 Under Keel Clearance Management).

1.3 (U) Document Overview

This manual provides users with practical guidance on downloading, installing, configuring, and operating the ShoreECDIS software. It covers dataset management, map display navigation, pick report execution, route planning, hazard detection, and route simulation.

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2. (U) Referenced Documents

(U) Table 2.1 Referenced Documents

(U) Document Number, Revision/Date	(U) Document Title
IHO S-100 Standard	IHO Universal Hydrographic Data Model (v5.0 / v5.2)
ShoreECDIS Ref Doc v2.0.0	ShoreECDIS Reference Document & Specification Matrix

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3. (U) Software Summary

3.1 (U) Software Application

ShoreECDIS is designed as a demonstration testbed application. Key operational capabilities include:

- Loading and displaying S-100 hydrographic datasets and exchange sets.
- Dynamic portrayal binding using Feature Catalogs (FC) and Portrayal Catalogs (PC).
- Detailed feature pick reporting and debugging inspection tools.
- Interactive route planning with automated route checking against safety contours and hazards.
- Route simulation (manual and automated) and real-time route monitoring.

Note: ShoreECDIS is intended for operational demonstration of S-100 ECDIS capabilities. It is **not** intended for formal data verification and validation—the companion **S100Viewer** app should be used for data verification and validation tasks.

3.2 (U) Software Inventory

The software distribution includes the following primary application packages:

- Shore ECDIS x64.msi (or version-equivalent installer)
- Installed Feature Catalogs (.xml) and Portrayal Catalogs (.zip/folder schema)

3.3 (U) Software Environment

To deploy and run ShoreECDIS, the target system requires:

- **Operating System:** Microsoft Windows 10/11 (64-bit).
- **Hardware:** Standard PC workstation with graphics display supporting hardware acceleration, mouse with scroll wheel, and keyboard.
- **Prerequisites:** Users must uninstall any versions of ShoreECDIS less than version v2.0.0 before installing a new release.
- **SENC Compatibility:** Beginning in version 2.1.0, the SENC database location has been moved so that older versions of the software remain fully usable alongside new installations.

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3.4 (U) Software Organization and Overview of Operation

Current Support Specifications (ShoreECDIS Version 2.1.0)

Product	Supported Feature Catalog (FC) Versions	Supported Portrayal Catalog (PC) Versions
S-101 (Electronic Navigational Chart)	• 1.0.2 • 1.1.0 • 1.2.3 • 1.5.0 • 2.0.0	• 1.0.2 • 1.1.0 • 1.2.3 • 1.5.0 • 2.0.0
S-102 (Bathymetric Surface)	• 2.0.0 • 2.1.2 • 2.2.0 • 3.0.0	• 2.0.0 • 3.0.0
S-104 (Water Level Information)	• 2.0.0	N/A
S-111 (Surface Currents)	• 1.0.0 • 1.2.0 • 2.0.0	• 1.0.0 • 2.0.0
S-124 (Navigational Warnings)	• 1.0.0 • 2.0.0	• 1.0.0 • 2.0.0
S-129 (Under Keel Clearance Management)	• 1.0.0 • 1.1.0 • 2.0.0	• 1.0.0 • 1.1.0 • 2.0.0

Limitations

- Feature catalogs must conform strictly to S-100 5.0 or 5.2 schemas.
- Portrayal catalogs and Dataset encodings must conform to S-100 5.2 schemas.
- Datasets must match the corresponding installed Feature Catalog version.

- Older datasets and catalogs outside specified schemas are unsupported and may cause the application to crash.

3.5 (U) Contingencies and Alternate States and Modes of Operation

If an unsupported or corrupted dataset/catalog is loaded, ShoreECDIS may terminate unexpectedly. Users should restart the application, clear problematic datasets from the Chart Library, and re-import valid datasets conforming to supported S-100 schema versions.

3.6 (U) Security, Licensing and Privacy

ShoreECDIS is distributed as standard testbed software under U.S. Navy / NIWC-LANT guidelines. Distribution and use must conform to applicable Navy and IHO licensing guidelines.

3.7 (U) Assistance and Problem Reporting

For assistance, standard queries, software updates, or defect reporting, contact the development testbed team:

- **Email:** NIWCTestbed@us.navy.mil
- **IHO Repository:** IHO Geospatial Information Registry
- **NIWC Repository:** GitHub

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4. (U) Access to Software

4.1 (U) First Time User of the Software

(U) Access Control

ShoreECDIS does not enforce user login or role-based account restrictions. Access is determined by local workstation permissions.

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(U) Installation and Setup

1. **Uninstall Prior Versions:** Ensure any earlier version than v2.0.0 of ShoreECDIS is completely uninstalled prior to proceeding.

2. **Download Applications:**

   - Navigate to the IHO Geospatial Information Registry or NIWC Github Repository
   - Open the **Repository** section and select **Item 5** (NIWC) S100Viewer and Shore ECDIS).
   - Download *NIWC_ShoreECDIS2.1_20260810XX.zip*

3. **Execute Installers:** Unpack the archive and execute the installation *Shore ECDIS x64.msi*

4. **Datasets and Catalogs:** Download necessary S-100 datasets, S-100 Feature Catalogs Portrayal Catalogs from the IHO GitHub repository as needed.

4.2 (U) Initiating a Session

Launch ShoreECDIS from the Windows Start menu. Pin it to start or your application taskbar for fast access.

The main navigation interface will open displaying the Chart View if data sets were previously installed, or migrated from v2.0. The application will open to the Chart Library if no datasets have been installed yet.

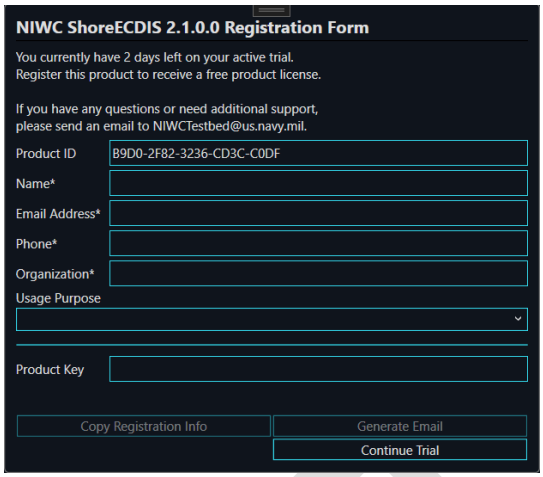
4.3 (U) License Key Registration

Upon launching the application for the very first time, a registration pop-up dialog box will appear requiring user input before software activation can be finalized:

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1. **Product ID Generation:** The dialog displays a hardware-specific **Product ID** automatically generated for your device.

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2. **Required User Details:** The registration form prompts the user to enter the following information:
    - **Name**
    - **Email Address**
    - **Phone Number**
    - **Organization**
    - **Usage Purpose**
3. **Trial Period:** Users are granted a **3-day trial period** upon initial launch, during which they can continue to use the software without restrictions while license processing takes place
4. **Requesting and Entering the License Key:**
    - The application prompts the user to submit the requested information to the testbed team at NIWCTestbed@us.navy.mil.
    - Upon receipt, the NIWC Testbed team will issue an official product license key.
    - Enter the received product license key into the registration window to complete software registration and unlock full application access beyond the trial period.
5. **Product Key Email Response:**
    - When the email response is received, copy the text on the line after "Product Key:"
    - Paste into "Product Key" field of the registration form.

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5. (U) Processing Reference Guide

5.1 (U) Capabilities

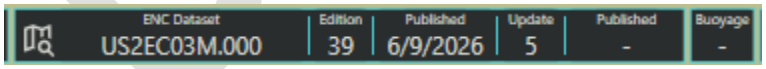
ShoreECDIS provides a comprehensive S-100 visualization environment with the following functional workflows:

- **Dataset Management:** Ingestion and automated catalog association.
- **Map Rendering & Inspection:** Scale zooming, pick reports, and feature debugging.
- **Voyage Planning:** Route creation, waypoint entry, and hazard analysis.
- **Navigation Simulation:** Automated/manual ship simulation and route monitoring.
- **GPS Configuration:** Configuration for Serial, TCP, or UDP source of NMEA0183 messages for the following message types: $GPRMC, $GPVTG, $HETHS, $HEHDG, $HEHDM, $HEHDT. Alternative to simulated routes can come from GPS equipment simulator.

5.2 (U) Conventions

- **Tooltips:** Hovering over any UI control presents contextual guidance.
- **Symbolization Icons:**

    - Magnifying glass icon in the legend indicates that higher-resolution data is available under the cursor. (Bottom of chart view)



    - Bug icon (Include debugging information) toggles expanded metadata display. (Pick report window)

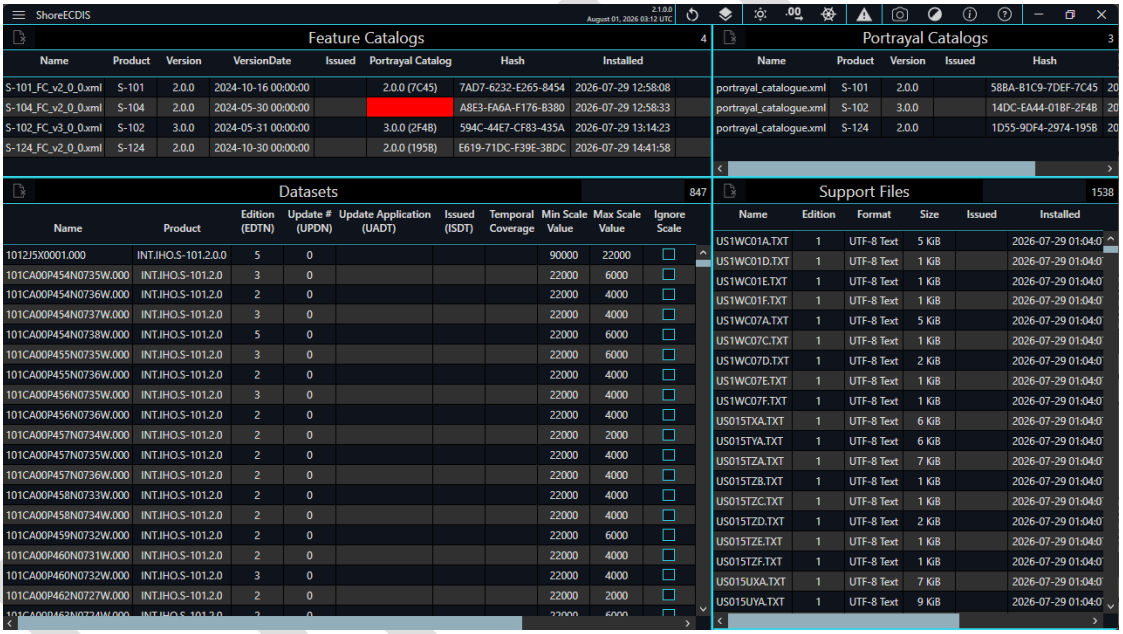
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5.3 (U) Processing Procedures

5.3.1 Ingesting & Installing Datasets

Datasets and catalogs can be imported directly using drag-and-drop actions.

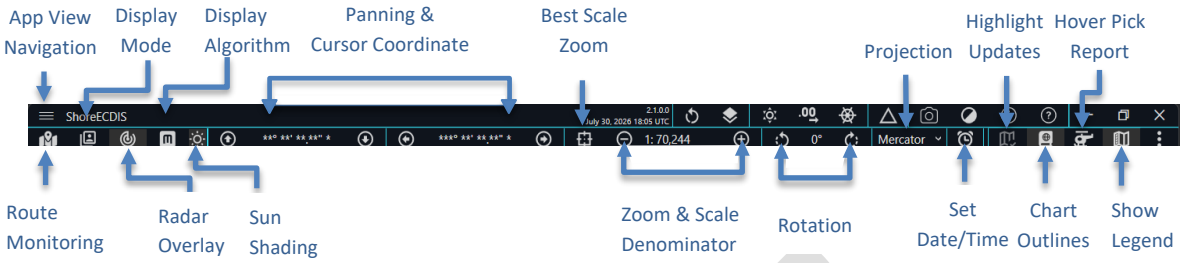
5. Open ShoreECDIS and navigate to either the **Chart View** window or the **Chart Library** window.
6. **Feature Catalogs First:** Feature catalogs (.xml) **must** be installed prior to loading related datasets.
7. Loading Methods:



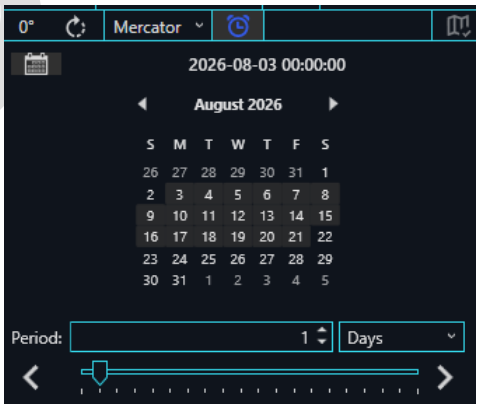
- **Individual File:** Drag and drop a single dataset or catalog file into the app window.
- **Exchange Set:** Drag and drop an entire CATALOG.031 or CATALOG.XML exchange catalog file.
- **Folders:** Drag and drop an entire folder containing dataset files. Loose files that are not datasets will be interpreted as support files.

8. ShoreECDIS will automatically associate loaded datasets with their appropriate Feature Catalog and Portrayal Catalog. The latest catalogs with a matching major and minor version number to the dataset will be chosen, (See 5.4)

5.3.2 Displaying Datasets & Map Navigation



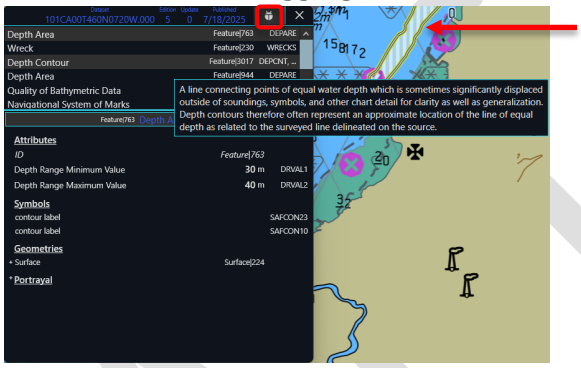
- **Jump to Dataset:** Double-click a dataset entry in the **Chart Library** menu to immediately center and zoom to its location, as well as changing the time setting if the dataset is temporal in nature.
- **Outline Zoom:** Zoom in directly on the outline box of a dataset in the Chart window to load and render it.
- **Zooming:**
  - Use the mouse wheel forward/backward.
  - Press + or - on the keyboard.
  - Use top bar magnification icons
  - Choose the best scale for charts at ownship's position (Best Scale Zoom button)
- **Re-center View:** Double-click any blank area in the chart window to re-center the chart display.
- **Set Time:** Double clicking the clock icon will change the time display mode from current to custom. If Custom time is chosen, the icon looks like like a normal clock, and single click will open a calander date range selector, and time segment picker. If the current time mode is selected the clock will have a checkbox instead of the right angle hands as pictured below.



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5.3.3 Feature Pick Report & Debugging

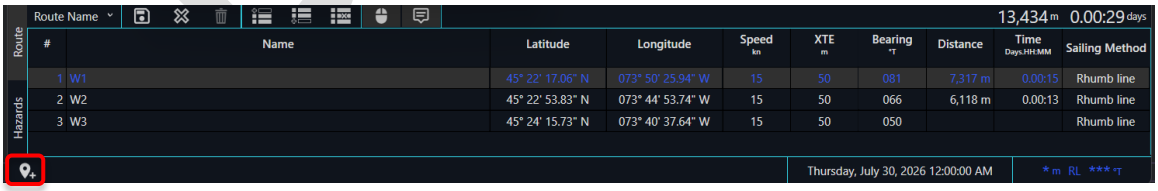
9. **Perform Pick Report:** Click on any displayed feature on the chart window to open the Feature Detail View. Hover over the feature to see it's feature catalog description.
10. **Review Feature Metadata:** View dataset edition, update number, publication date, feature class (DEPARE, M_COVR, M_SDAT, etc.), depth ranges, and symbol codes.
11. Enable Debugging Mode:



- Click the **Bug icon** (Include debugging information) at the top right of the pick report window.
- When enabled, additional technical attributes, viewing groups, drawing priorities, and display planes are displayed.
- Hovering the cursor over a feature row in the pick report highlights the corresponding geometry on the chart.

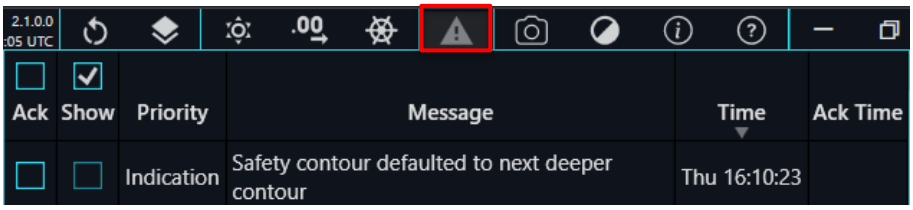
5.3.4 Route Planning & Hazard Detection

12. **Open Route Planner:** Select the **Route Planner** icon from the bottom left.



13. **Draft Route:** Place waypoints on the chart window to construct route legs.
14. **Save Route:** Enter a route name and click the **Save Route** button (or press Ctrl+S).

15. Automated Safety & Hazard Check:



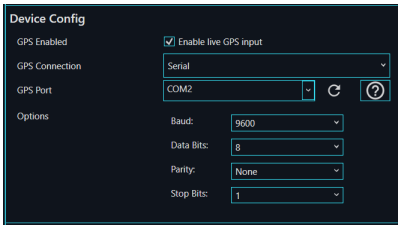
- ShoreECDIS automatically evaluates the route against underlying bathymetry and safety contours.
- Select the **Hazards** tab to review detected alerts (e.g., *Crossing safety contour*).
- Click a hazard row entry to highlight it on the chart; double-click the row to zoom directly to the hazard location.

5.3.5 Route Simulation & Monitoring

16. **Activate Route Simulator:** Select a saved route from the Route Simulator dropdown menu. The vessel symbol will move automatically to a position near the first waypoint.



17. **Manual Simulation:** Alternatively, use the **Manual Simulator** controls to adjust speed (kts) and heading (XTD, Look-ahead time in minutes).
18. **GPS Simulation:** Open "Settings" from **App View Navigation** (hamburger menu) in top left.
    Find "Device Config" and choose a suitable GPS configuration for your simulator or equipment.



Choose "Live GPS" from Route Simulator dropdown. If GPS signal fails, you may

choose “Manual Simulator” to input a speed and direction for dead reckoning.

19. **Track Vessel:** Click the **Route Monitoring** button on the top status bar to actively track the ship's position and progress along the route. The chart will center on the ship’s position once clicked, and when it nears the edge of the chart display.

5.4 (U) Related Processing

- **Automated Catalog Binding:** ShoreECDIS automatically scans installed Feature Catalogs and Portrayal Catalogs to match incoming dataset header attributes without requiring manual re-linking. Datasets with version 2.0.0 could be portrayed and interpreted by feature and portrayal catalogs of 2.0.0 to 2.0.99+, if such catalogs were installed. As soon as the new catalog is available, the best version will be chosen to make chart portrayal images.

5.5 (U) Data Backup

Saved routes (.rtz files) and user settings are preserved in the user’s local application directory `%LOCALAPPDATA%\NIWC\ShoreECDIS\Routes`. Users can back up route files by copying them from the application data directory.

5.6 (U) Recovery from Errors, Malfunctions, and Emergencies

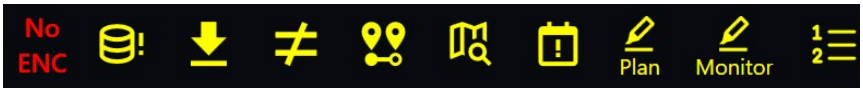
If the application crashes due to an incompatible dataset:

1. Restart ShoreECDIS. If it crashes immediately, first try deleting `%LOCALAPPDATA%\NIWC\ShoreECDIS\ShoreECDIS.user.config`
2. Open the **Chart Library** from “App View Navigation” (hamburger menu).
3. Remove the conflicting dataset or feature catalog.
4. Verify that datasets match supported S-100 schema versions (S-100 v5.0 / v5.2) before reloading.

5.7 (U) Messages

- **Crossing safety contour:** Indicates a planned route leg intersects water shallower than the configured safety contour value.

- **Better data available under cursor:** Indicated by a magnifying glass legend icon when higher-resolution charts cover the current view area.
  - **Possible Warnings at bottom of chart:**



    - **No ENC:** No S-101 or S-57 Data Set Found in view
    - **All Datasets Visible:** Chart Library data sets exist in view, but aren't being portrayed
    - Safety Downgraded: Specified safety contour not present, next deeper contour selected
    - **Different Reference System:** Coordinate reference system from data set not used
    - **Manual Position Adjustment:** Not used
    - **Larger Scale ENC Available:** A more accurate chart is available at a more zoomed scale
    - **Display Not Real Time:** The Current date isn't being used in chart render pipeline
    - **Route Plan Unsafe:** Off state of route planning across safety contour, prohibited areas and hazards
    - **Monitored Route Unsafe:** Off state of safety contour, prohibited area and hazard indication in route monitoring
    - **Out of Sequence Update:** Data set loaded with missing update between base and currently visible

5.8 (U) Quick-Reference Guide

Action	Shortcut / Gesture
Zoom In / Out	Mouse wheel or + / - keys
Re-Center Chart	Double-click chart window
Jump to Dataset	Double-click dataset in Chart Library
Pick Report	Right-click feature on chart
Save Route	Ctrl+S or Save icon in Route Planner

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Highlight Hazard	Click entry in Hazards tab

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6. (U) Notes

(U) Acronyms and Abbreviations

- **ECDIS:** Electronic Chart Display and Information System
- **ENC:** Electronic Navigational Chart
- **FC:** Feature Catalog
- **IHO:** International Hydrographic Organization
- **NIWC:** Naval Information Warfare Center
- **PC:** Portrayal Catalog
- **SUM:** Software User Manual
- **XTD:** Cross Track Distance

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7. (U) Appendices

(No separate appendices required for initial software release.)

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